Hades Model Report

v0.8.22 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Subalkaline**
Submode	**Tholeiitic**
Data suite	**./ontong_java_plateau.xlsx**
FeO/Fe^{2+}	**0.9**
Stages Run	Primary Primitive

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 48.17 wt.%

	Primary	Primitive	Total
Olivine	100.0	18.35	67.51
Spinel	0.0	5.42	2.16
Clinopyroxene		24.7	9.83
Plagioclase		51.53	20.51
FC%	29.0	27.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	0.95	0.95	0.95	0.95	0.95
Clinopyroxene	0.25	0.25	0.25	0.25	0.25
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Sample	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary	1187-12 primary	46.86	0.54	11.01	11.14	19.89	9.04	1.29	0.22
Primitive	1187-12	49.52	0.75	15.47	11.12	8.36	12.64	1.82	0.32
Basic	807-12	50.85	1.13	14.0	12.74	7.15	11.9	2.15	0.08
Intermediate									
Evolved									
End									

Table 3: Major oxide compositions of the model phases (wt.%).

Computed Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary	46.86	0.54	11.01	11.14	19.89	9.04	1.29	0.22
Primitive	49.45	0.74	15.23	11.19	8.84	12.44	1.79	0.31
Basic	50.85	0.98	14.03	12.68	7.15	11.91	1.99	0.42
Intermediate								
Evolved								
End								

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Primary > Primitive

PRIMARY

Population size	50
Iterations	415/500
Cost function	0.36
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	46.86	49.52	49.45	0.14
TiO2	0.54	0.75	0.74	1.53
Al2O3	11.01	15.47	15.23	1.6
Fe2O3	11.14	11.12	11.19	0.69
MgO	19.89	8.36	8.84	5.74
CaO	9.04	12.64	12.44	1.53
Na2O	1.29	1.82	1.79	1.57
K2O	0.22	0.32	0.31	1.57

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	28.0
Olivine	100.0
Spinel	0.0
Clinopyroxene	0.0
Orthopyroxene	0.0
Plagioclase	0.0
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	0.0
Ilmenite	0.0

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	39.52	0.02	0.1	14.48	45.53	0.36	0.0	0.0

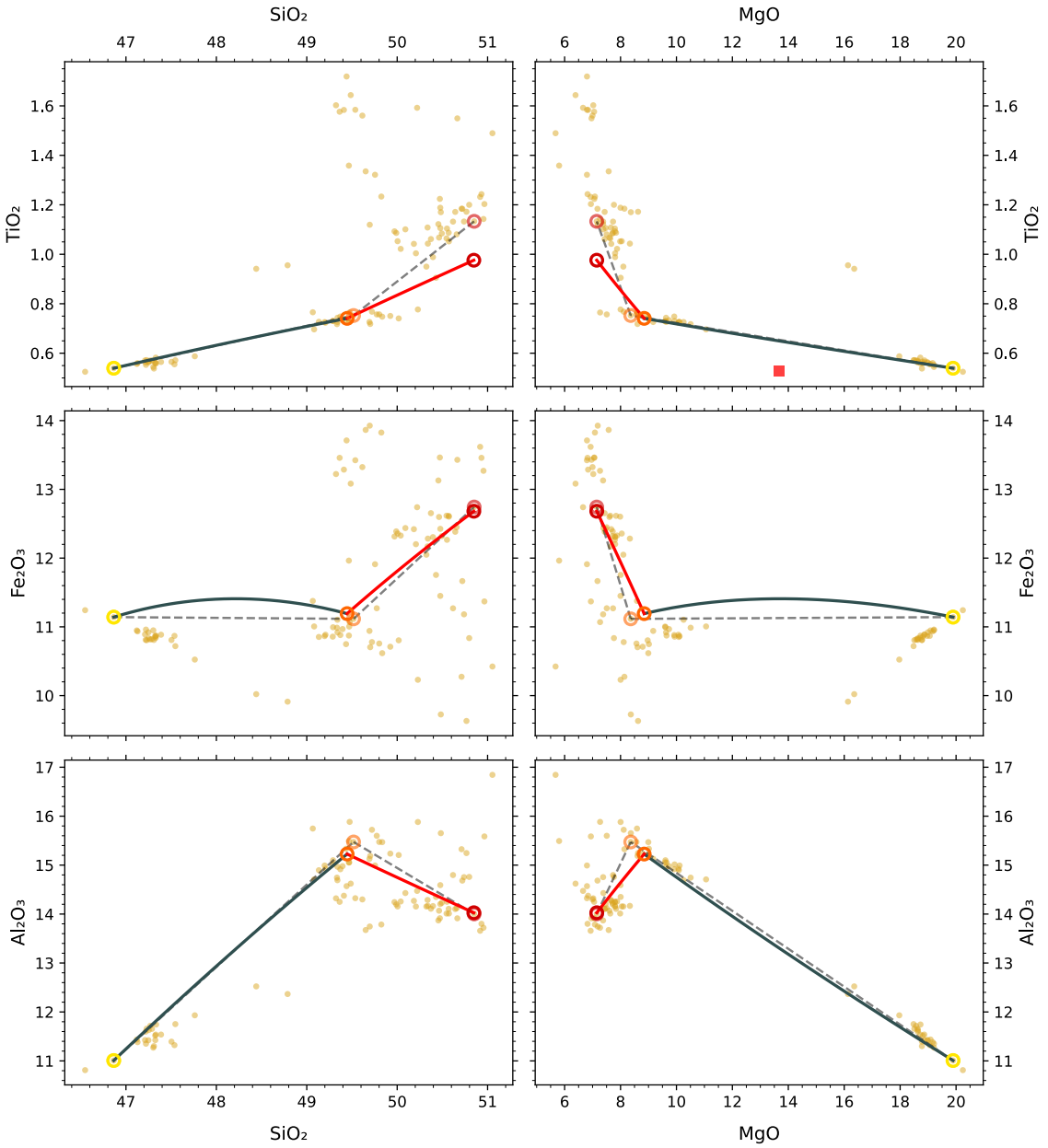
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	39.52	0.02	0.1	14.48	45.53	0.36	0.0	0.0
spinel	0.26	0.53	30.33	30.52	13.68	24.34	0.34	0.0
clinopyroxene								
orthopyroxene								
plagioclase								
amphibole								
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

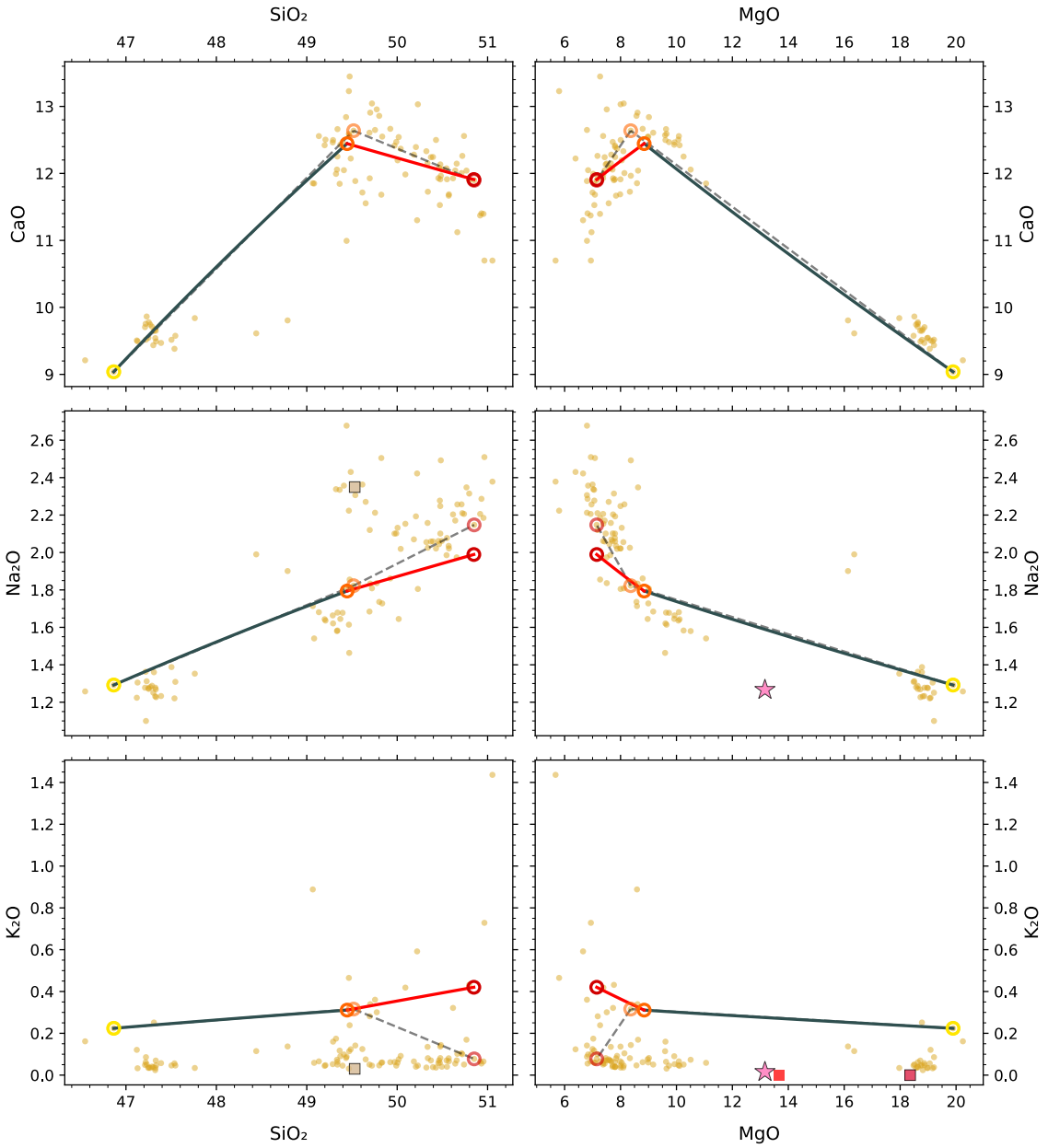


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primary to Primitive transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Primitive > Basic

PRIMITIVE

Population size	**50**
Iterations	**500/500**
Cost function	**1.51**
Trace analysis?	**False**

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	49.45	50.85	50.85	0.01
TiO2	0.74	1.13	0.98	13.87

	Initial	Expected	Computed	% Error
Al2O3	15.23	14.0	14.03	0.17
Fe2O3	11.19	12.74	12.68	0.48
MgO	8.84	7.15	7.15	0.02
CaO	12.44	11.9	11.91	0.1
Na2O	1.79	2.15	1.99	7.36
K2O	0.31	0.08	0.42	441.39

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	27.0
Olivine	18.35
Spinel	5.42
Clinopyroxene	24.7
Orthopyroxene	0.0
Plagioclase	51.53
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	0.0
Ilmenite	0.0

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	45.59	0.11	18.47	7.5	13.16	13.9	1.27	0.02

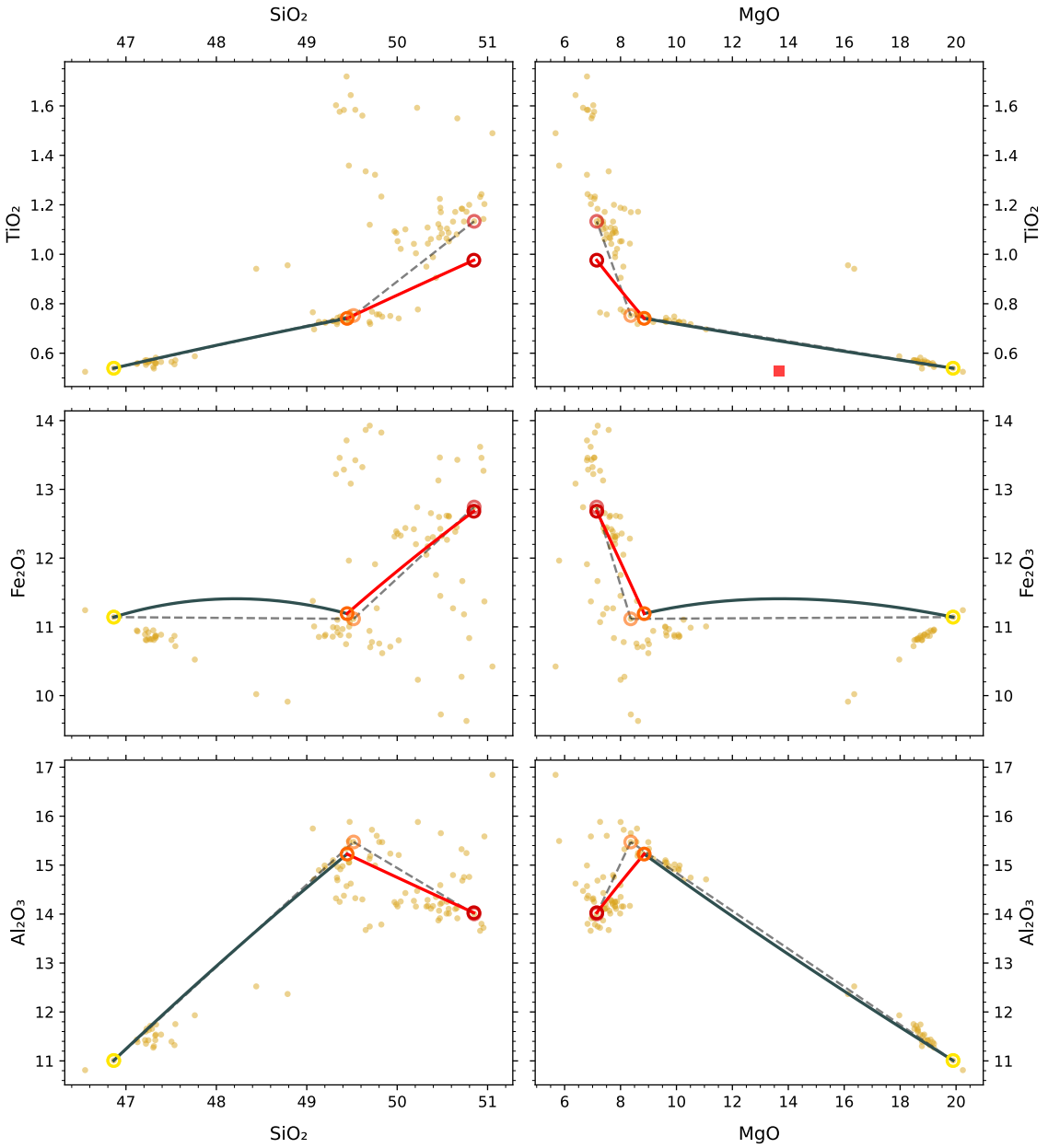
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	38.59	0.03	0.08	18.87	42.03	0.4	0.0	0.0
spinel	0.26	0.53	30.33	30.52	13.68	24.34	0.34	0.0
clinopyroxene	52.5	0.29	3.14	7.95	18.36	17.62	0.15	0.0
orthopyroxene								
plagioclase	49.53	0.0	31.11	0.82	0.34	15.82	2.35	0.03
amphibole								
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 12: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

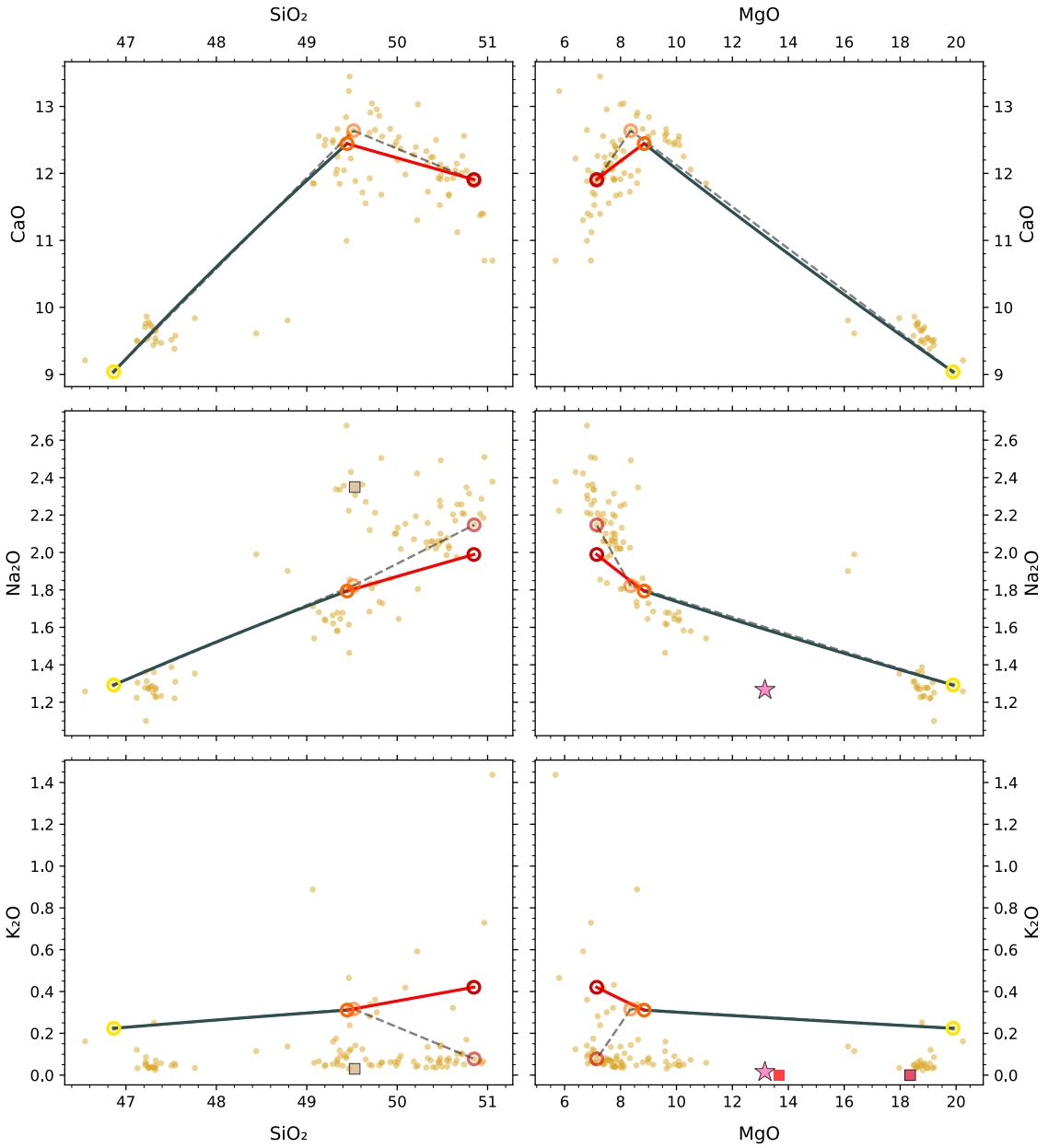


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

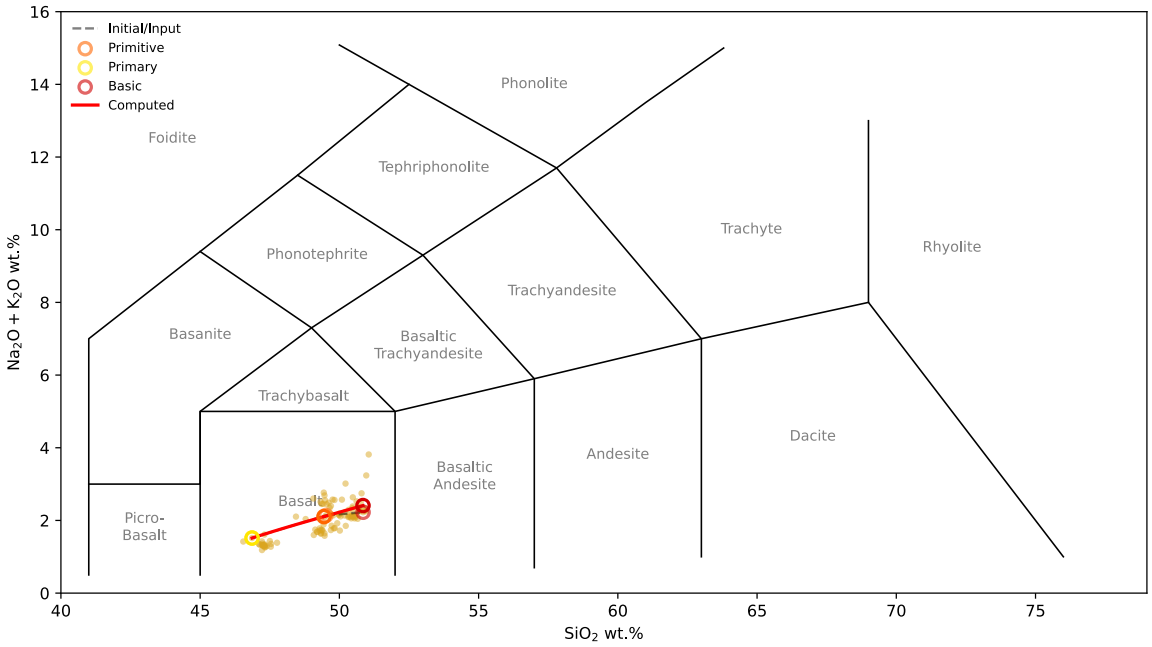


Figure 3: Total Alkali-Silica (TAS) diagram of the model phases.

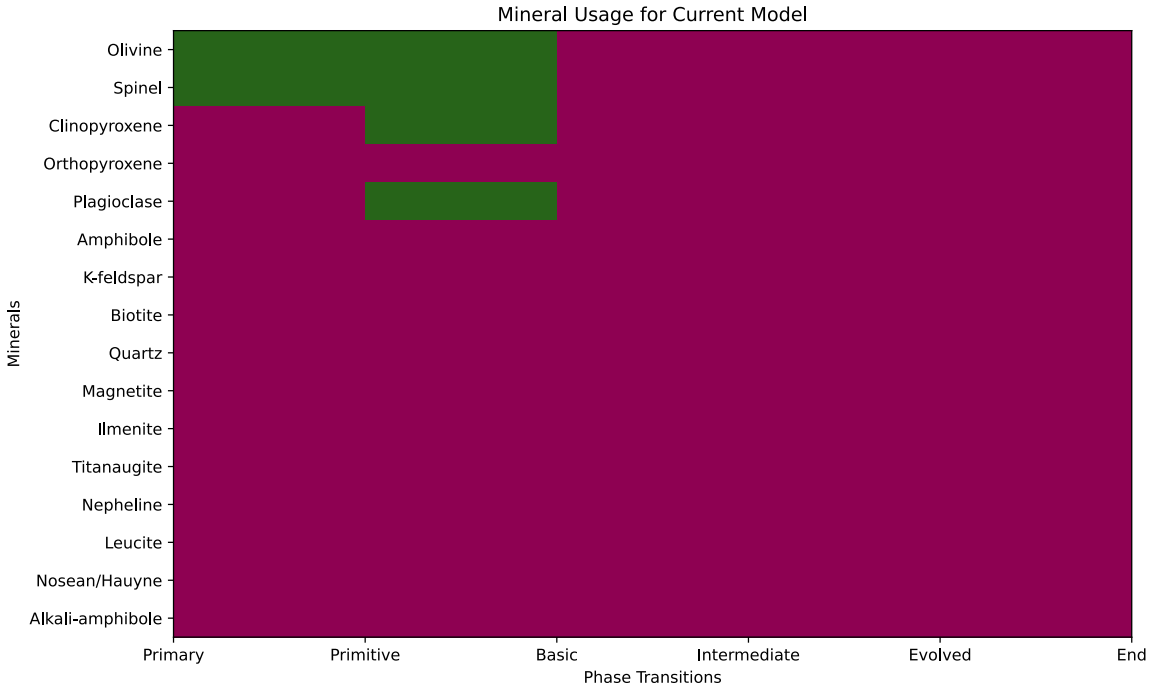


Figure 4: Mineral usage across the model phases.

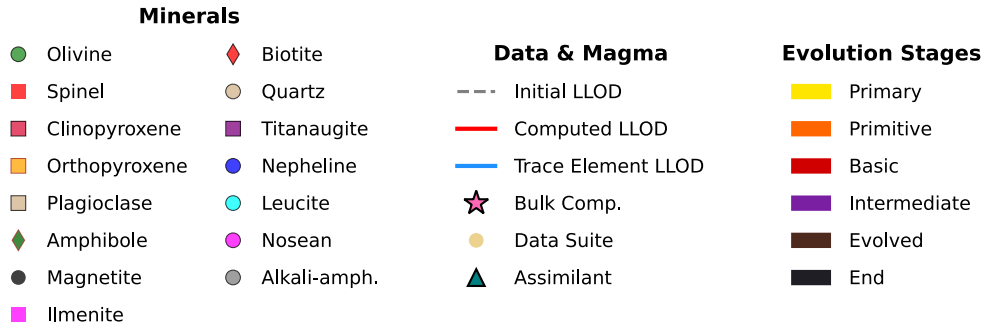


Figure 5: Legend for mineral symbols used in the figures.